REGULARITY OF FORMAL FIBRES FOR A SMOOTH ALGEBRA OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Regularity of a local ring and its preservation under etale maps and localisation. used at proof lines 13276-13320. Carries the property from the polynomial case to the smooth case.
- Structure of a smooth algebra over a field as a localisation of an etale algebra over a polynomial ring. used at proof lines 13300-13320. Reduces the statement to the polynomial case.
- The completion of a polynomial ring at a maximal ideal is a power series ring with as many variables as generators. used at proof lines 13320-13332. Is the explicit computation that supplies regularity in the remaining case.

Source reference. Original lemma 0BIZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14507–14507. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 030R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3958–3958. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 037X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4216–4216. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1062–1062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4049–4049. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00TT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1087–1087. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0353: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4114–4114. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0C2X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15172–15172. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Fix $n \geq 0$. Properties (A), (B), (C), (D), and (E) hold for $P(k \to R) =$"$R \otimes_k k'$ has (R_n) for all finite extensions k'/k".*

Proof. Let $k \to R$ be a ring map where k is a field and R a Noetherian ring. Assume $P(k \to R)$ is true. Let K/k be a finitely generated field extension. By Algebra, Lemma 030R we can find a diagram

$$\begin{array}{ccc} K & \longrightarrow & K' \\ \uparrow & & \uparrow \\ k & \longrightarrow & k' \end{array}$$

where k'/k, K'/K are finite purely inseparable field extensions such that K'/k' is separable. By Algebra, Lemma 037X there exists a smooth k'-algebra B such that K' is the fraction field of B. Now we can argue as follows: Step 1: $R \otimes_k k'$ satisfies (R_n) because we assumed P for $k \to R$. Step 2: $R \otimes_k k' \to R \otimes_k k' \otimes_{k'} B$ is a smooth ring map (Algebra, Lemma 00T4) and we conclude $R \otimes_k k' \otimes_{k'} B$ satisfies (R_n) by Algebra, Lemma 033A (and using Algebra, Lemma 00TT to see that the hypotheses are satisfied). Step 3. $R \otimes_k k' \otimes_{k'} K' = R \otimes_k K'$ satisfies (R_n) as it is a localization of a ring having (R_n). Step 4. Finally $R \otimes_k K$ satisfies (R_n) by descent of (R_n) along the faithfully flat ring map $K \otimes_k R \to K' \otimes_k R$ (Algebra, Lemma 0353). This proves (A). Part (B) follows too: a Noetherian ring has (R_n) if and only if all of its local rings have (R_n). Part (C). This follows from Algebra, Lemma 033A as the fibres of a regular homomorphism are regular (small detail omitted). Part (D). This follows from Algebra, Lemma 0353 (small detail omitted).

Part (E). Let l/k be a separable algebraic extension of fields and let $l \to R$ be a ring map with R Noetherian. Assume that $k \to R$ has P. We have to show that $l \to R$ has P. Let l'/l be a finite extension. First observe that there exists a finite subextension $l/m/k$ and a finite extension m'/m such that $l' = l \otimes_m m'$. Then $R \otimes_l l' = R \otimes_m m'$. Hence it suffices to prove that $m \to R$ has property P, i.e., we

may assume that l/k is finite. If l/k is finite, then l'/k is finite and we see that

$$l' \otimes_l R = (l' \otimes_k R) \otimes_{l \otimes_k l} l$$

is a localization (by Algebra, Lemma 0C2X) of the Noetherian ring $l' \otimes_k R$ which has property (R_n) by assumption P for $k \to R$. This proves that $l' \otimes_l R$ has property (R_n) as desired. □